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Review article

RNA modification-mediated aberrant alternative splicing: An underlying driver of disease pathogenesis

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ABSTRACT

RNA modification and alternative splicing are central mechanisms in the regulation of eukaryotic gene expression. RNA modifications, which occur in a variety of chemical forms, exert widespread influences on RNA structure, stability and function. Meanwhile, alternative splicing significantly contributes to proteomic diversity by generating multiple transcript isoforms from a single gene. Dysregulation of RNA modifications can disrupt the normal splicing process, leading to the production of aberrant splice variants that are associated with a range of human diseases. In this review, we summarize the current understanding of how common RNA modifications regulate alternative splicing, and discuss recent advances in the study of diseases arising from aberrant alternative splicing caused by abnormal RNA modifications. A deeper mechanistic understanding of these processes provides a foundation for the development of novel diagnostic and therapeutic strategies. These include the design of small-molecule modulators targeting specific RNA modification sites or splicing factors; the precise correction of aberrant modifications and splicing events, and the application of gene editing technologies to regulate the expression of key splicing factors, all of which hold promise for future clinical interventions.

1. Background and significance

RNA modification and alternative splicing are prominent research areas in molecular biology that play crucial roles in the regulation of gene expression and are closely associated with the development of numerous human diseases. The rapid development of high-throughput sequencing technologies has significantly enhanced our understanding of these post-transcriptional regulatory mechanisms, opening new avenues for elucidating the molecular pathogenesis of various disorders.

RNA modification refers to the covalent chemical alterations of RNA bases and sugar moieties at specific positions within RNA molecules. These modifications exhibit diverse chemical forms and are widely present in various types of RNA, including messenger RNAs (mRNAs), transfer RNAs (tRNAs), ribosomal RNAs (rRNAs), small nuclear RNAs (snRNAs), small nucleolar RNAs (snoRNAs), microRNAs (miRNAs), and long non-coding RNAs (lncRNAs). To date, over 140 distinct types of RNA modifications have been identified across archaea, bacteria, viruses and eukaryotes. Commonly studied modifications include N^6-methyladenosine (m^6A), 5-methylcytosine (m^5C), N^1-methyladenosine (m^1A), N^7-methylguanosine (m^7G), 2′-O-methylation (2′-O-Me), and pseudouridine (Ψ), among others [1]. These modifications can profoundly influence RNA stability, translation efficiency, structural conformation, and interactions with other biomolecules, thereby enabling precise regulation of gene expression at the post-transcriptional level. Numerous studies have demonstrated that dysregulation of RNA modifications is closely linked to the onset and progression of various human diseases. Among these, m^6A modifications have been most extensively studied and are clearly implicated in cancer and neurological diseases. Aberrant m^6A levels may contribute to disease pathogenesis by altering the expression and processing of key regulatory genes [2−4].

Alternative splicing is a fundamental process wherein precursor mRNAs (pre-mRNAs) generated through transcription are differently spliced to produce multiple mature mRNA isoforms. This mechanism significantly enhances the complexity of the proteome and the diversity of biological functions, serving as a critical post-transcriptional regulatory strategy in organisms. Over 90 % of human genes undergo alternative splicing, yielding protein isoforms with distinct functionalities that are essential for maintaining normal cellular physiology by selectively including or excluding exons or retaining introns. However, aberrant alternative splicing can lead to alterations or loss of protein function, contributing to various diseases. A number of studies of myelodysplastic syndromes (MDS) revealed that mutations

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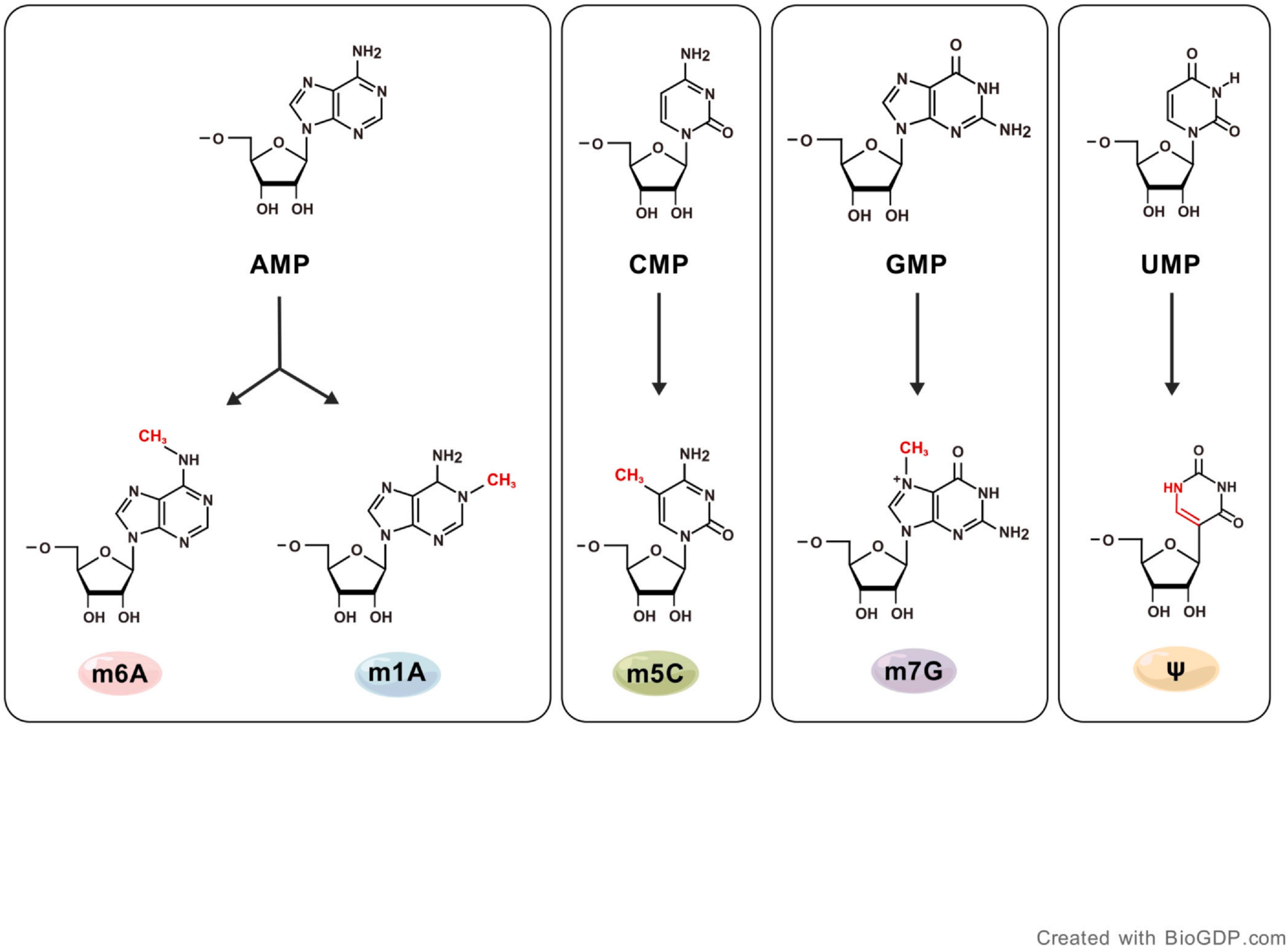

Fig. 1. Chemical structures and modification sites of major RNA modifications. Schematic representations of the base structures of four canonical nucleoside monophosphates—adenosine monophosphate (AMP), cytidine monophosphate (CMP), guanosine monophosphate (GMP), and uridine monophosphate (UMP)—and their corresponding modified derivatives: N^6-methyladenosine (m^6A), N^1-methyladenosine (m^1A), 5-methylcytidine (m^5C), N^7-methylguanosine (m^7G), and pseudouridine (Ψ). Each panel displays the unmodified nucleoside structure at the top, with an arrow indicating the formation of the modified nucleoside at the bottom. Chemical modification sites are highlighted in red to emphasize structural changes. Figure created using BioGDP.com. [258].

in SF3B1, which encodes a component of RNA spliceosome complex, were highly associated with a subtype characterized by ring sideroblastic anemia [5−8]. In the 2016 revised edition of the WHO classification criteria, SF3B1 mutations have been included as one of the diagnostic criteria for MDS-RS [9]. These mutations may disrupt the binding of splicing factors to pre-mRNA, resulting in the formation of dysfunctional spliceosomes that produce abnormal proteins, thereby triggering disease symptoms. Research by Yong Zhen Xu's group at the Shanghai Institutes for Biological Sciences demonstrated that SF3B1 mutations (e.g., K700E, R625C/H) reduced its interaction strength with the ATPase Prp5, impacting the accuracy of intron recognition and elucidating the molecular mechanisms underlying SF3B1 mutation-induced splicing defects [10]. In addition to the spliceosomal mis-function caused by SF3B1 mutations, the abundance of wild-type SF3B1 protein itself dictates splicing fidelity and disease phenotype, and this dosage control is fine-tuned by m^6A modification. Mechanistically, within bone-marrow cells of MDS patients, ALKBH5 removes m^6A at position A88 in the 5′ UTR of SF3B1 mRNA, thereby relieving translational repression and elevating SF3B1 protein levels by approximately 1.4-fold [11]. Intriguingly, this increase in wild-type SF3B1 exerts a protective effect in early-stage MDS by preserving splice-site accuracy. However, in the setting of cooperative high-risk mutations, it can synergistically accelerate progression to acute myeloid leukemia [11,12].

The interplay between RNA modifications and alternative splicing is intricate. On one hand, RNA modifications can directly influence the binding affinity of splicing factors to pre-mRNAs; for example, N^2-methylguanosine (m^2G) on U6 snRNA promotes efficient pre-mRNA splicing [13]. Similarly, m^6A modifications can regulate exon inclusion or exclusion by recruiting specific RNA-binding proteins and affecting their interactions with splice site sequence. Specifically, the m^6A reader YTHDC1 facilitates exon inclusion by recruiting SRSF3 to m^6A sites, while its recruitment of SRSF10 promotes exon skipping [14]. However, alternative splicing events can also impact the distribution and function of RNA modifications. Changes in splicing patterns may alter regions of RNA containing specific modification sites, thereby influencing overall RNA modification levels and functional outcomes. This interconnected regulatory network enhances the precision and complexity of gene expression regulation but also complicates the investigation of its role in disease mechanisms.

An in-depth exploration of the mechanisms by which RNA modifications induce aberrant alternative splicing in diseases holds immense medical significance. Firstly, it aids in better understanding the molecular underpinnings of disease development. Many diseases, including cancers, neurodegenerative diseases, and autoimmune diseases, remain incompletely understood at the molecular level. By uncovering associations between RNA modification and alternative splicing abnormalities in these diseases, novel targets and strategies for diagnosis and treatment can be identified. For example, identifying key links between tumor-specific alternative splicing changes caused by abnormal RNA modifications could facilitate the development of targeted inhibitors against abnormal splicing events or related modifying enzymes, offering new avenues for precision oncology. Secondly, this research is crucial for biomarker discovery and drug development. Abnormal RNA modifications and alternative splicing events often appear early stage in disease progression and exhibit specificity and stability, making them promising candidates for early diagnostic biomarkers. Moreover, leveraging a deeper understanding of RNA modification and alternative splicing regulatory mechanisms allows for the design and development of more effective and specific therapeutic molecules aimed at modulating these aberrant molecular events, thus providing potential interventions for disease management.

2. Well-characterized RNA modifications

A variety of RNA modifications exist in eukaryotes, with emerging evidence highlighting their distinct biological functions, regulatory mechanisms, and roles in modulating alternative splicing. Among the most extensively studied are N^6-methyladenosine, 5-methylcytosine, N^1-methyladenosine, N^7-methylguanosine, and pseudouridine. These chemical modifications are shown in Fig. 1, and their related writers, erasers, and readers are shown in Table 1.

2.1. m^6A modification

m^6A is formed by methylation at the sixth nitrogen position of adenosine resides (Fig. 1). This epitranscriptomic mark is widely present in eukaryotic mRNAs, lncRNAs, as well as certain snRNAs and tRNAs. The deposition of m^6A exhibits sequence dependency, with a strong preference for the RRACH motif, where R represents A or G, and H represents A, C, or U. Within mRNAs, m^6A modifications are particularly enriched in the 3′ untranslated regions (UTRs) and in proximity to the stop codon [15].

The methyltransferase complex (MTC), responsible for catalyzing N^6-adenylate methylation is composed of three core components - METTL3, METTL14 and WTAP - as well as four regulatory subunits: VIRMA, ZC3H13, HAKAI, and RBM15 (Table 1). Among these, METTL3 possesses the primary catalytical activity, while METTL14 plays a crucial role in recognizing the RNA substrate and facilitating METTL3-mediated methylation. WTAP is essential for the proper localization and functional integrity of the MTC complex. In terms of regulatory mechanisms, m^6A levels are dynamically controlled by both writer and eraser enzymes. m^6A modifications can be reversed by demethylases, including FTO and ALKBH5, which remove methyl groups from N^6-methyladenosine, thereby modulating m^6A abundance in a reversible and dynamic manner (Table 1) [16,17].

m^6A plays diverse and critical roles in the regulation of gene expression. One of its well-characterized functions is the promotion of RNA degradation, primarily mediated by members of the YTHDF protein family. For example, YTHDF2 facilitates the decay of m^6A-containing transcripts by interacting with at least three major RNA degradation pathways: the CCR4-NOT de-adenylation complex, the HRSP12-RNase P/MRP endonuclease pathway, and the UPF1-DCP1A decapping complex [18]. In contrast, IGF2BPs recognize m^6A modification via their KH domains and stabilize target RNA, thereby counteracting the destabilizing effects of YTHDF proteins [19]. In addition to regulating mRNA stability, m^6A also modulates translation efficiency of mRNAs [20]. Methylation within the CDS region has been shown to enhance translation by recruiting YTHDC2, an RNA helicase that unwinds secondary structures and facilitates ribosome scanning [21]. However, not all m^6A sites promote translation; certain sites are bound by YBX3, which suppresses translational initiation and reduces the expression of the corresponding transcripts (Table 1) [22].

More importantly, m^6A and its readers play a pivotal role in splicing control. First, experimental studies have revealed that depletion of writer complexes such as METTL3/METTL14 leads to widespread exon skipping and intron retention, indicating that the deposition of m^6A and its coordinated recognition by readers are essential for maintaining normal splicing profiles [23]. Notably, m^6A modifications facilitate the formation of a nuclear "m^6A switch" through the recruitment of specific reader proteins, thereby profoundly influencing alternative splicing outcomes. A well-characterized example is the nuclear reader YTHDC1, which, upon binding to m^6A-marked transcripts, preferentially recruits the pro-inclusion SR protein SRSF3 while excluding the skipping-promoting factor SRSF10, thus fine-tuning the exon inclusion/skipping balance [24]. Furthermore, the nuclear reader hnRNPG recognizes m^6A

Table 1
Writer, eraser, reader, and target RNA molecules associated with major RNA modifications.

Modifications	Writer	Eraser	Reader	Target RNA
m^6A	METTL3/14 WTAP VIRMA ZC3H13 HAKAI RBM15	FTO ALKBH5	YTHDFs YTHDC1/2 IGF2BP1 HNRNPC YBX3 hnRNPG	mRNA rRNA snRNA snoRNA miRNA lncRNA circRNA
m^7G	METTL1/WDR4 RNMT/RAM WBSCR22/TRMT112	N.A.	CBC (CBP80/20) eLF4E IGF2BP3 QKIs	mRNA rRNA tRNA snRNA snoRNA miRNA
m^5C	NSUN family (NOL1/NOP2/sun) DNMT2	TET	YBX1 YTHDF2 ALYREF	mRNA rRNA tRNA lncRNA miRNA
m^1A	TRMT6/61 A	FTO ALKBH3	YTHDC1 YTHDF2	mRNA rRNA tRNA
ψ	PUS1–10 TRUB1/2	N.A.	N.A.	mRNA rRNA tRNA snRNA

marks near splice sites in nascent pre-mRNA. By modulating the occupancy pattern of RNA Polymerase II (RNAPII) at these loci, this interaction promotes exon inclusion [25]. In disease contexts, such as lung cancer cells, the formation of the YTHDC1–hnRNPK–AURKA complex has been shown to promote skipping of specific exons (e.g., RBM4 Exon3), leading to the production of oncogenic splice variants (RBM4-S) and underscoring the pathophysiological relevance of m^{6}A-reader-mediated splicing regulation [26].

In summary, m^{6}A is a pivotal post-transcriptional modification that coordinately regulates RNA metabolism, including splicing, stability, and translation. The precise control exerted by this dynamic system is crucial for normal gene expression and its dysregulation is directly linked to diseases such as cancer.

2.2. m^{7}G modification

The m^{7}G modification, 7-methylguanosine, is a prevalent post-transcriptional RNA modification found in various RNA species, including mRNAs, rRNAs, and tRNAs (Fig. 1).

In mammals, m^{7}G modifications are regulated by several methyltransferases, including the METTL1/WDR4 complex, the RNMT/RAM complex, and WBSCR22 in association with its cofactor TRMT112 (Table 1). Among these METTL1, in complex with its regulatory partner WDR4, is a key enzyme responsible for catalyzing the methylation of the N^7 position of guanosine. This complex primarily mediates m^{7}G formation at position 46 of tRNA [27], creating a positively charged binding platform that enhances substrate specificity and facilitates tRNA maturation [28]. Recent studies have demonstrated that the METTL1/WDR4 complex also catalyzes internal m^{7}G formation within mRNAs [29]. The RNMT/RAM complex is primarily involved in the formation of the m^{7}G cap structure at 5′ end of mRNA transcripts, a critical step in mRNA processing and translation initiation [30]. In contrast, WBSCR22, together with its activating subunit TRMT112, functions as the methyltransferase responsible for introducing m^{7}G modifications into 18S rRNAs. This activity is crucial for pre-rRNA processing and the biogenesis of the 40S ribosomal subunit [31].

The m^{7}G modification plays diverse yet essential roles across various RNA species, contributing significantly to fundamental biological processes. In mRNAs, m^{7}G is most commonly found at the 5′ end as part of the 7-methylguanosine cap (m^{7}G cap), which is critical for mRNA stability and initiation of translation [32]. This modified cap is specifically recognized by the cap-binding complex (CBC), composed of CBP80 and CBP20, which protects the mRNA from exonuclease-mediated degradation and enhances its structural integrity [33]. During translation initiation, the eukaryotic initiation factor 4E (eIF4E) first binds to the m^{7}G cap, facilitating the recruitment of additional translation initiation factors and the small ribosomal subunits (Table 1). This sequential assembly leads to the formation of the translation initiation complex, a key step in protein synthesis [34,35]. Additionally, mRNA internal m^{7}G is specifically recognized by the reader protein IGF2BP3. In contrast to IGF2BP1, which binds m^{6}A to stabilize target transcripts, IGF2BP3 preferentially associates with m^{7}G-modified mRNAs and accelerates their degradation [29]. Curiously, the Quaking proteins (QKIs) also serve as specific readers of mRNA internal m^{7}G. Among the isoforms, QKI7 selectively shuttles m^{7}G-modified transcripts into stress granules (SGs) via a direct C-terminal interaction with the SG core component G3BP1, thereby governing the stability and translational efficiency of target mRNAs under stress conditions [36]. In tRNA, m^{7}G is predominantly located within the anticodon loop, where it contributes to the stabilization and proper folding of this functionally critical region [37]. This structural optimization enhances the fidelity of codon recognition during translation, thereby reducing the likelihood of mispairing between the tRNA and mRNA codons and ensuring accurate protein synthesis [38]. Moreover, m^{7}G-modified tRNAs exhibit increased efficiency in ribosome binding and amino acid delivery, promoting overall translational efficiency [31]. Within rRNAs, m^{7}G modifications are implicated in ribosome biogenesis and function. These modifications influence the interactions between rRNAs and ribosomal proteins, aiding in the formation and stabilization of specific three-dimensional structures. As a result, m^{7}G plays a pivotal role in the proper assembly and functional competence of the ribosome [39].

2.3. m^{5}C modification

m^{5}C is an RNA modification generated by the addition of a methyl group to the 5th carbon position of cytosine. This modification is widely present across various types of RNA molecules, including mRNA, tRNA, and rRNA (Fig. 1) [40].

The NOL1/NOP2/sun (NSUN) family and DNA methyltransferase 2 (DNMT2) are the primary methyltransferases responsible for catalyzing the formation of m^{5}C in RNA (Table 1) [41]. Among these, NSUN2 is considered the most critical enzyme, mediating m^{5}C modification in both mRNAs and tRNAs by transferring a methyl group from S-adenosylmethionine (SAM) to the 5th carbon of cytosine, thereby generating m^{5}C [42]. NSUN3 specifically introduces m^{5}C modification within the anticodon stem-loop (ASL) and TΨC loop of mitochondrial tRNAs. This modification is essential for maintaining the tertiary structure of tRNAs and enhancing their ability to accurately base-pair with mRNA codons [43−45]. In contrast, NSUN6 primarily targets the CTCCA motif in the 3′UTR region of mRNA, exerting its effects by influencing mRNA translation termination and stability [40,46−48]. In addition to writers, eraser enzymes also play a role in the dynamic regulation of m^{5}C. The ten-eleven translocation (TET) family of dioxygenases is known to oxidize m^{5}C, leading to its eventual demethylation and restoration of unmodified cytosine [41,49]. However, compared to the relatively well-characterized m^{5}C writing machinery, the mechanisms underlying m^{5}C erasure remain less understood and require further investigation. Recognition of m^{5}C-modified RNA is mediated by specific reader proteins, among which Y-box binding protein 1 (YBX1) has been extensively studied [49]. YBX1 exhibits enhanced binding affinity toward RNA sequences containing specific m^{5}C motifs such as "CA(A/C)C", further contributing to the regulation of gene expression at the post-transcriptional level [50,51]. A key tryptophan residue (Trp45 and Trp65) within the cold-shock domain (CSD) of YBX1 serves as the core recognition site for m^{5}C, engaging in hydrophobic interactions that enable selective binding to modified RNA (Table 1) [51,52]. This interaction allows YBX1 to recruit mRNA-stabilizing factors such as ELAVL1 (HuR) and PABPC1, thereby enhancing the stability of m^{5}C-modified transcript [53]. Additionally, YTHDF2 has been identified as a reader of m^{5}C modification. On mRNA, YTHDF2 binds m^{5}C-marked transcripts and recruits PABPC1 to enhance their stability [54]. Concurrently, YTHDF2 recognizes m^{5}C residues on ribosomal RNA, thereby facilitating pre-rRNA cleavage and promoting ribosomal RNA maturation [55].

The m^{5}C modification has been shown to influence RNA secondary structure and stability, thereby affecting the binding affinity and specificity of splicing factors and other RNA-binding proteins. By altering RNA conformation, m^{5}C can modulate key post-transcriptional regulatory processes, including splicing and transcript localization. Notably, m^{5}C and m^{6}A modifications exhibit cooperative regulatory effects in the expression of certain genes, such as p21. For instance, NSUN2-mediated m^{5}C methylation, in conjunction with METTL3/METTL14-catalyzed m^{6}A modification, enhances the translational efficiency of p21 mRNA, contributing to the induction of cellular senescence [56].

Unlike more dynamic RNA modifications such as m^{6}A, which can be rapidly installed or removed in response to cellular signals, m^{5}C is chemically stable and maintains a relatively persistent methylation state. This characteristic allows m^{5}C to exert long-term regulatory effects on RNA function. In contrast, the reversible nature of m^{6}A enable it to dynamically influence alternative splicing patterns, and its dysregulation has been associated with various pathological conditions.

2.4. m^1A modification

m^1A is an evolutionarily conserved RNA modification formed by the addition of a methyl group to the N^1 position of adenosine. This modification is predominantly found in structurally and functionally critical regions across different RNA species, including the 5′ UTR and near the start codon of mRNAs, specific nucleotide positions within transfer RNAs tRNAs, and the small subunit of rRNAs (Fig. 1). The presence of m^1A is dynamically regulated by specific writer and eraser enzymes, enabling it to modulate key biological processes such as mRNA translation and RNA stability. Dysregulation of m^1A modification has been implicated in various pathological conditions [57].

In mRNAs, m^1A modifications are particularly enriched in the vicinity of the 5′ UTR and the start codon [58]. Within the 5′ UTR, m^1A modifications can influence translation initiation by altering local RNA secondary structures, for instance, by promoting the formation of stem-loop structures that hinder scanning of ribosomal small subunits, thereby reducing translational efficiency [58]. In the coding sequence, m^1A may interfere with codon-anticodon base pairing during translation elongation, leading to ribosomal pausing or mistranslation events, which can affect protein output and fidelity [59]. Intriguingly, YTHDF2 functions not only as an m^6A reader but also as a reader of m^1A-modified mRNAs, promoting their degradation and thereby exerting a critical role in post-transcriptional gene regulation [60]. Furthermore, m^1A modification at transcription start sites of both rRNA and mRNA has been shown to stabilize RNA structures, facilitating ribosome assembly and mRNA maturation [61,62]. These modifications contribute to the proper folding and functional competence of ribosomes, ultimately enhancing global translation efficiency [63].

In tRNAs, m^1A modifications are primarily located at evolutionarily conserved positions, with the most well-characterized being m^1A at position 58 (m^1A_{58}) within the anticodon loop. This modification is catalyzed by the tRNA methyltransferase complex composed of TRMT6 and TRMT61A (Table 1) [60,64]. m^1A_{58} plays a crucial role in maintaining tRNA tertiary structure under stress conditions such as high temperature or elevated salt concentrations, preventing structural unraveling or misfolding [65]. By stabilizing the overall conformation of tRNA, m^1A ensures accurate recognition and binding of the amino acid acceptor arm to its corresponding aminoacyl-tRNA synthetase. Moreover, similar to m^7G, m^1A in the anticodon loop enhances the stability of tRNA-mRNA base pairing during decoding, thereby minimizing mismatch errors and ensuring the fidelity of the aminoacylation reaction [66].

2.5. Ψ modification

Pseudouridine, Ψ, is a modified nucleotide formed in RNA through the isomerization of uridine (U) at the C5-N1 glycosidic bond (Fig. 1). The formation of Ψ is mediated by two major mechanisms: an RNA-dependent pathway and an RNA-independent pathway.

The RNA-dependent pathway involves H/ACA box snoRNAs and their associated protein factors, including Cbf5/NAP57 (the catalytic subunit), Nhp2, Gar1, and Nop10. These components form a ribonucleoprotein complex that guides site-specific pseudouridylation. In contrast, the RNA-independent pathway relies on pseudouridine synthases (PUS), a family of enzymes capable of directly catalyzing Ψ formation without the need for guide RNAs. To date, thirteen human PUS enzymes (e.g., PUS1–PUS10) have been identified (Table 1) [67]. Notably, no specific eraser enzyme has yet been identified that can reverse this modification, suggesting that Ψ may function as a stable, irreversible mark under physiological conditions.

Ψ modification is ubiquitously present across various RNA types. In eukaryotic tRNA, Ψ is most commonly found at position 55 within the Thymine-Pseudouridine-Cytosine (TΨC) loop - a highly conserved site critical for tRNA structure and function. This modification is primarily catalyzed by TRUB1/TRUB2 and plays an essential in maintaining tRNA stability (Table 1) [68]. Moreover, Ψ modifications within the anticodon stem-loop enhance decoding fidelity and are evolutionarily conserved from bacteria to eukaryotes [68]. In rRNAs, Ψ modification influence ribosome conformation and facilitate the cooperative interaction between the large and small ribosomal subunits, thereby ensuring efficient and accurate translation [69−71]. Within mRNAs, Ψ sites exhibit distinct positional preferences: approximately 60 % are enriched in the 3′ UTR, potentially influencing microRNA binding or transcript stability [72]; about 35 % are located within the coding sequences, with some overlapping near stop codons, where they may regulate readthrough translation events [72,73].

In pre-mRNA, Ψ is significantly enriched in proximal introns, alternatively spliced segments, and splice-factor-binding sites, implying a regulatory role in splicing control [74]. Notably, Ψ modifications are also found at snRNAs and play pivotal roles in RNA splicing. Within the branch-point-recognition region of U2 snRNA, every uridine is quantitatively converted to Ψ; this modification strengthens base-pairing between U2 snRNA and the pre-mRNA substrate, thereby enhancing the spliceosome's ability to recognize divergent branch-point sequences [75,76]. Meanwhile, Ψ enhances U2 snRNA binding to pre-mRNAs via interactions with the Prp5 ATPase [77,78].

Collectively, these findings indicate that Ψ can modulate the splicing process. Nevertheless, the pathological consequences of altered Ψ patterns—particularly in relation to aberrant alternative splicing—remain largely unexplored. Future studies should integrate multi-omics approaches with functional assays to comprehensively characterize the spatiotemporal dynamics of Ψ modifications and explore their potential as therapeutic targets.

3. Mechanisms and regulation of alternative pre-mRNA splicing

Gene transcription yields precursor mRNA, which typically consists of multiple exons interspersed with intronic sequences. Prior to translation, the pre-mRNA must undergo splicing—a process in which introns are removed and exons are ligated to generate mature mRNA capable of directing protein synthesis and participating in essential cellular functions. A key mechanism that significantly expands the coding capacity of the genome is alternative splicing; whereby different combinations of exons are selected during the splicing process. This generates multiple mature mRNA isoforms from a single pre-mRNA transcript, which can subsequently be translated into functionally distinct protein variants. Through this mechanism, alternative splicing enables the generation of proteomic diversity from a finite set of genetic instructions.

At the heart of this process lies the spliceosome, the molecular machinery responsible for catalyzing pre-mRNA splicing in eukaryotes. The spliceosome performs the critical tasks of recognizing splice donor and acceptor sites, excising introns, and joining adjacent exons. It is a large and dynamic ribonucleoprotein complex composed primarily of small nuclear ribonucleoproteins (snRNPs) and numerous associated auxiliary proteins. The assembly and activation of the spliceosome represent key regulatory steps in the splicing pathway. Its dynamic behavior, along with its interactions with cis-regulatory elements and trans-acting factors, ultimately determines the patterns of alternative splicing and contributes to the regulation of gene expression.

3.1. Trans-acting splicing regulatory factors

The spliceosome, the molecular machinery responsible for pre-mRNA splicing, is primarily composed of snRNPs, which serve as core functional components. These include U1, U2, U4, U5, and U6 snRNPs, each playing distinct and essential roles during the splicing process. For instance, U1 snRNP recognizes and binds to the 5′ splice site of pre-mRNA, whereas U2 snRNP interacts with the branch point sequence near the 3′ splice site, facilitating the formation of the early spliceosomal complex [79,80]. The U4/U6 and U5 snRNPs initially associate as a tri-snRNP complex. During spliceosome activation, U4 snRNA dissociates from U6 snRNA, allowing U6 to interact with U2 snRNA and promote the structural rearrangements necessary for catalytic activity [81−83]. Subsequently, U5 snRNP helps position the flanking exons in close proximity, forming the intronic loop structure. This configuration enables the two sequential transesterification reactions that result in intron excision and exon ligation [84].

Beyond these core snRNP components, the spliceosome also incorporates a variety of non-snRNP protein factors that regulate spliceosome assembly, stability, and function. Among these are the serine/arginine-rich (SR) proteins, a family of RNA-binding splicing regulators that play pivotal roles in both constitutive and alternative splicing. SR proteins typically contain one or more RNA recognition motifs (RRMs) and a characteristic serine/arginine-rich (RS) domain. During alternative splicing, they can bind to exonic splicing enhancers (ESEs) to promote exon inclusion [85], or interact with intronic splicing silencers (ISSs) to inhibit splice site recognition and suppress exon inclusion [86], thereby modulating splice site selection. In addition to SR proteins, several accessory proteins contribute to the proper assembly and function of the spliceosome. These proteins assist in the localization and integration of snRNPs into active complexes, ensuring the fidelity and efficiency of the splicing reaction. A notable example is Prp8, a large and evolutionarily conserved protein that plays a central role in U5 snRNP organization and 5′ splice site recognition [86,87].

Another major class of trans-acting regulatory factors includes the heterogeneous nuclear ribonucleoproteins (hnRNPs). These proteins bind to various cis-regulatory elements within pre-mRNA, including splice sites, ESEs, and exonic splicing silencers (ESSs). Depending on the context and binding location, hnRNPs can either enhance or repress splicing events, often functioning antagonistically or cooperatively with SR proteins to fine-tune alternative splicing outcomes [88−90].

3.2. cis-acting splicing regulatory elements

Similar to transcriptional regulation, RNA splicing is governed by both trans-acting regulatory factors and cis-acting RNA sequence elements that collectively determine splicing outcomes. At the core of the splicing process are the fundamental splice sites—specific nucleotide sequences that demarcate exon–intron boundaries and serve as recognition signals for the spliceosome.

The 5′ splice site (5'SS) is typically located at the 5′ end of an intron and is characterized by a highly conserved "GU" dinucleotide, whereas the 3′ splice site (3′SS) resides at the 3′ end of the intron and is commonly defined by an "AG" consensus sequence [91−93]. These sequences are essential for spliceosomal recognition and catalytic cleavage during splicing, serving as primary determinants for both constitutive and alternative splicing events. Within the intron, the branch point (BP) lies upstream of the 3′SS and contains a conserved adenosine residue. This site forms a covalent lariat structure with the 5′SS guanine during the first step of splicing, playing a crucial role in intron excision and exon joining [94].

Beyond these canonical splice site sequences, a variety of cis-regulatory elements modulate alternative splicing by influencing the recruitment and activity of trans-acting splicing factors. These include:

- ESEs and ESSs, which are located within exons. ESEs promote spliceosome assembly and enhance exon inclusion in the mature mRNA by facilitating the binding of positive regulators such as SR proteins. In contrast, ESSs recruit splicing repressors like hnRNPs, increasing the likelihood of exon skipping [95−97].
- ISEs, which reside within introns, facilitate efficient intron removal by enhancing the interaction between spliceosomal components and splice sites, while ISSs inhibit splice site recognition, thereby suppressing intron excision and exon inclusion [98].

A number of studies have employed computational prediction tools or high-throughput screening approaches to identify functional cis-regulatory elements in specific genes, shedding light on the molecular mechanisms underlying splicing regulation [99−102].

In summary, alternative splicing is orchestrated by the coordinated interplay between the spliceosome—the central molecular machinery—and a network of cis-acting regulatory sequences and trans-acting RNA-binding proteins. Together, they influence splice site selection and exon combination patterns, enabling a single pre-mRNA transcript to give rise to multiple mature mRNA isoforms. This diversity translates into a wide array of protein isoforms with distinct functional properties, significantly expanding the complexity and adaptability of the proteome. Such regulatory flexibility allows gene expression to be finely tuned in response to tissue-specific and developmental cues.

3.3. Splice site recognition and spliceosome assembly

The process of pre-mRNA splicing begins with the precise recognition of splice sites by core spliceosomal components. The U1 small nuclear ribonucleoprotein (U1 snRNP) initiates spliceosome assembly by binding to the 5′SS. Concurrently, the heterodimeric U2 auxiliary factor (U2AF) facilitates 3′SS recognition: its U2AF65 subunit binds to the polypyrimidine tract (PPT) located upstream of the 3′SS, while U2AF35 specifically interacts with the conserved AG dinucleotide at the 3′SS. Together, these interactions ensure accurate recognition of the 3′SS and promote subsequent steps in spliceosome formation [103−106]. In parallel, various splicing regulatory factors—including SR proteins, hnRNPs, and the branch point-binding protein SF1/BBP—are recruited to the vicinity of the branch point sequence [107]. These proteins recognize specific cis-regulatory RNA elements through their RNA-binding domains, thereby establishing the early splicing complex (E complex). This complex promotes initial interactions between splice sites and facilitates the spatial approximation of exons, marking the transition from the E complex to the prespliceosomal A complex (Fig. 2) [99,108].

The next major step involves the recruitment of the U4/U6·U5 tri-snRNP complex. Within this complex, U4 snRNA remains tightly associated with U6 snRNA, where it functions as a negative regulator to prevent premature activation of U6 catalytic activity [82]. The U5 snRNP serves as a central scaffold within the tri-snRNP, playing a crucial role in positioning the exons for ligation [109]. During spliceosome activation, RNA helicases such as Brr2 and Prp2—both possessing ATP-dependent unwinding activity—mediate extensive structural rearrangements. These include the dissociation of U1 snRNP from the 5′SS and the release of U4 snRNA, allowing U6 snRNA to base-pair with U2 snRNA and form the active catalytic core of the spliceosome [110−114]. Simultaneously, U5 snRNP interacts with both the 5′ and 3′ exons, effectively bridging the two exon ends to facilitate their alignment for ligation [115,116]. Through a series of dynamic conformational changes, the spliceosome transitions into its catalytically active state (B complex and beyond), wherein the splice sites are precisely positioned to undergo the two sequential transesterification reactions that define the splicing reaction. These events ensure efficient and accurate execution of alternative splicing, ultimately yielding diverse mature mRNA isoforms tailored to specific cellular contexts (Fig. 2).

3.4. The two-step transesterification reaction in pre-mRNA splicing

Once the catalytically active spliceosome is fully assembled, the splicing reaction is initiated through a two-step transesterification mechanism. The first step represents a highly coordinated molecular transformation within the spliceosomal catalytic core. In this stage, the U2 small nuclear ribonucleoprotein utilizes ATP hydrolysis to drive structural rearrangements and facilitate base pairing between its own RNA component and the branch point sequence within the intron. This interaction results in the precise positioning and exposure of the conserved adenine residue at the branch site. This nucleophilic 2′-hydroxyl group then attacks the phosphodiester bond at the 5′SS, leading to its cleavage [117,118]. Consequently, a covalent lariat structure is formed

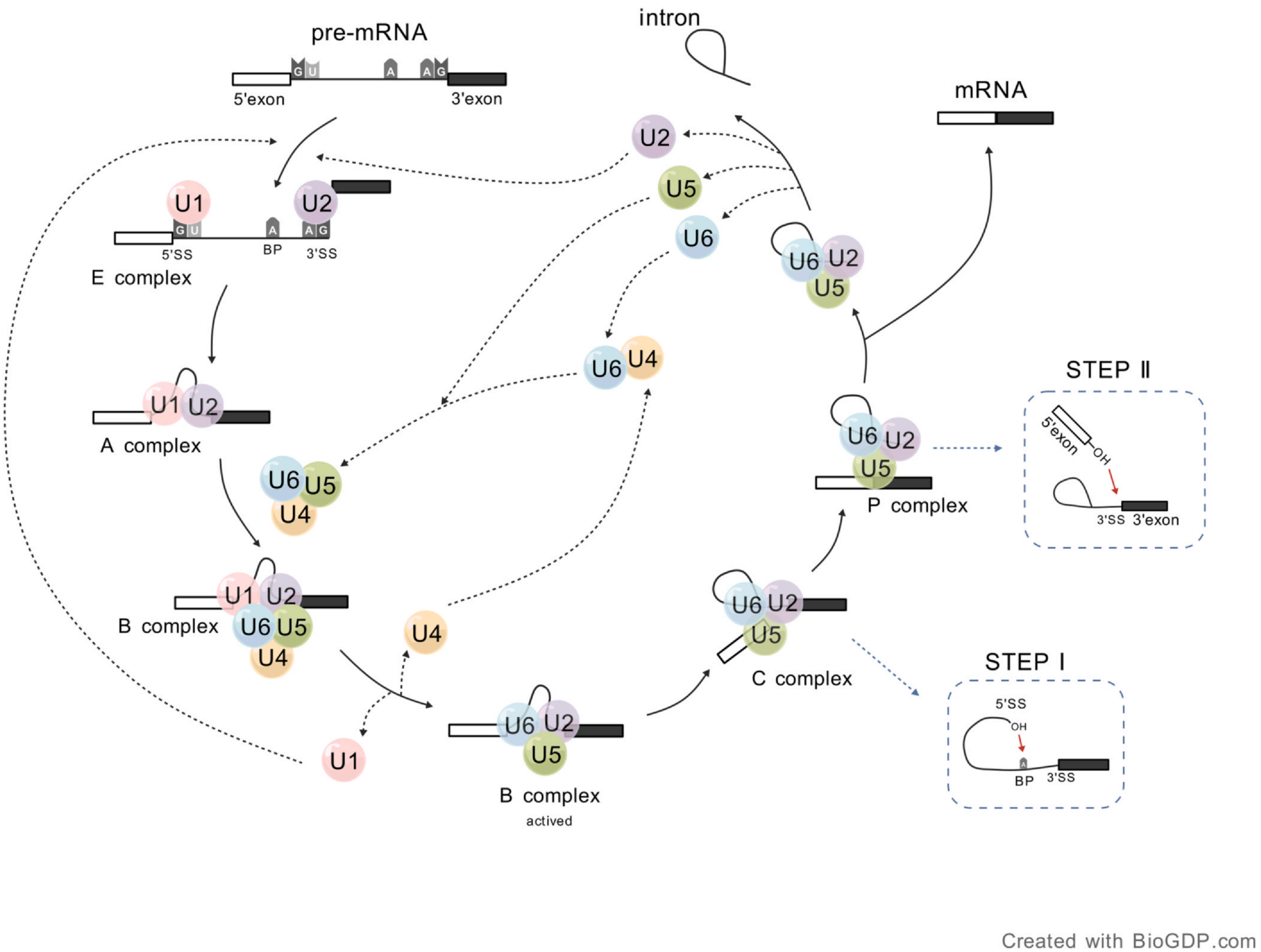

Fig. 2. Schematic representation of the canonical pre-mRNA splicing pathway in eukaryotes. This diagram illustrates the stepwise assembly and remodeling of the spliceosome during constitutive splicing of precursor mRNA (pre-mRNA). The process initiates with the recognition of the 5′ splice site (5′SS) by U1 small nuclear ribonucleoprotein (snRNP) and the binding of U2 snRNP to the branch point sequence (BPS), forming the early (E) complex. Subsequent juxtaposition of the 5′SS and 3′ splice site (3′SS) leads to the formation of the pre-catalytic A complex. Recruitment of the U4/U6·U5 tri-snRNP integrates into the complex, generating the B complex. Conformational rearrangements then trigger the release of U4 snRNP, allowing U6 snRNP to engage with U2 snRNP and form the catalytically active center. Dashed arrows indicate the association and dissociation of snRNP components during spliceosome maturation. The U2–U6 RNA helix serves as the catalytic core that mediates the two sequential transesterification reactions. Following the first transesterification, in which the 2′-hydroxyl group of the branch point adenosine nucleophilically attacks the 5′SS (indicated by a red arrow), the C complex is formed, and the intron adopts a lariat structure. In the subsequent step, the free 3′-OH of the 5′ exon attacks the 3′SS, resulting in exon ligation and intron excision during the transition to the post-catalytic P complex. Blue dashed boxes highlight the molecular details of the two transesterification reactions. Figure created using BioGDP.com. [258].

between the branch point adenosine and the guanine at the 5′SS [109], effectively transforming the linear pre-mRNA into a looped conformation. This lariat formation marks the transition from the B complex to the catalytically active C complex (Fig. 2), bringing critical components of the spliceosome into closer proximity and setting the stage for the second transesterification event.

In the subsequent step, the free 3′-hydroxyl group of the released 5′ exon performs a nucleophilic attack on the phosphodiester bond at the 3′SS, specifically at the conserved AG dinucleotide. This reaction results in the cleavage and reorganization of the phosphodiester backbone, culminating in the ligation of the two exons and the formation of the post-splicing P complex [119–122]. Concurrently, the excised lariat intron is efficiently released from the spliceosome. It is rapidly recognized and degraded by cellular nucleases, with Dbr1, an enzyme specialized in debranching lariat structures, serving as a key rate-limiting factor in this process [105,123]. This efficient degradation prevents the accumulation of non-functional RNA species and minimizes potential interference with gene expression, thereby contributing to the maintenance of cellular homeostasis. Upon completion of splicing, the mature mRNA undergoes further processing, including the addition of a 5′ cap and a 3′ poly(A) tail. The mature transcript is then exported from the nucleus via the nuclear pore complex, often in association with specific RNA-binding transport proteins. Once in the cytoplasm, it is directed to ribosomes where it serves as a template for protein synthesis during translation.

4. Association of RNA modifications with alternative splicing abnormalities in disease

4.1. RNA modification and alternative splicing abnormalities in cancer

A growing body of evidence indicates that aberrant alterations in RNA modifications frequently contribute to the dysregulation of alternative splicing. These disruptions in splicing patterns can profoundly influence key cellular processes, including tumor cell proliferation, apoptosis, invasion, and metastasis. As a result, such findings provide novel insights into the molecular mechanisms underlying cancer development and progression.

4.1.1. Lung cancer

Lung cancer (LC) is among the most prevalent and lethal malignant tumors worldwide, with high morbidity and mortality rates. It is broadly classified into two major pathological subtypes: non-small cell lung cancer (NSCLC) and small cell lung cancer (SCLC), with NSCLC accounting for approximately 85 % of all lung cancer cases.

Aberrant regulation of m^6A RNA modification has been frequently observed in NSCLC, playing a critical role in tumor development. METTL3, a key methyltransferase responsible for catalyzing m^6A modification, is significantly upregulated in NSCLC tissues compared to adjacent non-tumor tissues. This overexpression contributes to the activation of multiple oncogenes, including EGFR and KRAS [124,125]. Notably, METTL3 collaborates with YTHDC1, an m^6A reader protein, to promote the back-splicing of circIGF2BP3 and enhance its stability through m^6A-dependent mechanisms, resulting in its marked overexpression in NSCLC [126]. Following its nuclear export, circIGF2BP3 functions as a competitive endogenous RNA (ceRNA) by sponging miR-328–3p and miR-3173–5p. This interaction relieves the suppressive effects of these microRNAs on their target gene PKP3, thereby promoting its upregulation. PKP3 subsequently binds to FXR1, enhancing the stability of the deubiquitylating enzyme OTUB1. This leads to reduced ubiquitination and degradation of programmed death-ligand 1 (PD-L1), resulting in increased PD-L1 expression on the surface of tumor cells. Elevated PD-L1 levels facilitate its interaction with the PD-1 receptor on $CD8^+$ T cells, thereby suppressing anti-tumor immune responses and promoting immune evasion (Fig. 3) [127]. Targeting the circIGF2BP3/miRNA/PKP3 regulatory axis has been shown to significantly enhance the therapeutic efficacy of anti-PD-1 immunotherapy, highlighting its potential as a novel strategy for improving immunotherapeutic outcomes in NSCLC [126].

4.1.2. Breast cancer

Breast cancer (BC) is a malignant neoplasm arising from the uncontrolled proliferation of mammary epithelial cells. It is the most common cancer among women globally and is increasingly prevalent in China, where it accounts for approximately 7–10 % of all systemic malignancies. In major urban centers, BC has surpassed other cancers in incidence among women. Although its precise etiology remains unclear, risk factors such as hormonal imbalances, genetic predisposition, radiation exposure, and unhealthy lifestyle choices have been implicated. Importantly, emerging evidence highlights a significant role for aberrant RNA modifications and alternative splicing in the initiation and progression of breast cancer.

In breast cancer, m^6A RNA modification has emerged as a key epigenetic regulator involved in the expression and splicing of multiple oncogenic transcripts. METTL3, a core component of the m^6A methyltransferase complex, is frequently overexpressed in breast cancer tissues and is closely associated with tumor progression. METTL3-mediated m^6A modification stabilizes the long non-coding RNA—MALAT1, which, in turn, functions as a microRNA sponge by sequestering miR-26b. This derepresses the inhibitory effect of miR-26b on its target gene HMGA2, leading to increased HMGA2 expression. Elevated HMGA2 recruits histone methyltransferases such as EZH2 to the promoter region of E-cadherin, a key cell adhesion molecule, resulting in its transcriptional repression and loss of epithelial cell-cell adhesion. Concurrently, the expression of mesenchymal markers such as N-cadherin, Vimentin, and Fibronectin is upregulated, promoting epithelial-mesenchymal transition (EMT), tumor cell migration, and invasion. This intricate regulatory network illustrates the multilayered interplay between RNA methylation, non-coding RNA signaling, and transcriptional control, offering novel mechanistic insights into breast cancer metastasis and identifying potential therapeutic targets (Fig. 3) [128].

Moreover, METTL3 can modulate alternative splicing by depositing m^6A marks at splice site boundaries or through its enrichment at splice junctions of target transcripts, thereby altering the splicing patterns of critical genes and splicing factors [129]. Additionally, METTL3 has been shown to promote the translation of oncogenic proteins such as EGFR via an m^6A-independent mechanism, further contributing to tumor proliferation and metastasis [130].

4.1.3. Pancreatic cancer

Pancreatic cancer (PC) is often referred to as the "king of cancers" due to its aggressive nature, late diagnosis, and poor clinical outcomes. With the global aging population, its incidence has been steadily increasing [131,132]. In the United States, pancreatic cancer has become the third leading cause of cancer-related mortality [133]. Similarly, in China, the incidence is on the rise, and the overall 5-year survival rate for patients remains low, at approximately 10 % [134].

Chen et al. reported that CDC-like kinase 1 (CLK1) is significantly upregulated in pancreatic ductal adenocarcinoma (PDAC), the most common subtype of pancreatic cancer [135 – 138]. CLK1 plays a central role in the phosphorylation of splicing regulatory proteins. Functionally, high CLK1 expression promotes tumor cell proliferation and metastasis, as demonstrated in both in vitro and in vivo models. Mechanistically, CLK1 enhances the phosphorylation of SRSF5 at the Ser250 residue, promoting its binding to the pre-mRNA of METTL14, a key m^6A methyltransferase. This interaction leads to the retention of exon 10 in METTL14 transcripts, increasing its mRNA stability and consequently elevating global m^6A levels. This upregulation of m^6A modification contributes to enhanced metastatic potential in PDAC (Fig. 3) [139].

In addition, dysregulation of splicing factors has also been implicated in pancreatic cancer progression. CUGBP Elav-like family member 2 (CELF2), a key regulator of alternative splicing, is frequently downregulated in pancreatic cancer and is associated with poor prognosis. This reduction in CELF2 expression is partly attributed to the downregulation of the m^6A eraser ALKBH5, which results in elevated m^6A levels on CELF2 mRNA. The hyper-methylated CELF2 transcripts are specifically recognized and bound by the m^6A reader YTHDF2, leading to their accelerated degradation. Reduced CELF2 levels promote the expression of the standard isoform CD44s, which activates the Endoplasmic Reticulum-Associated Degradation (ERAD) pathway, thereby enhancing cancer cell proliferation, migration, and stemness. In contrast, CELF2 overexpression favors the production of the variant isoform CD44v, which suppresses the ERAD pathway and inhibits tumor progression (Fig. 3) [140].

4.1.4. Thyroid cancer

Thyroid cancer (TC) is a malignant tumor originating from the follicular epithelium or parafollicular epithelium of the thyroid gland. It is the most common cancer of the head and neck, with the incidence rate increasing at the highest rate among solid malignant tumors in the last 20 years. Its etiology is unknown, and it is related to oncogenes, heredity, and ionizing radiation. According to its pathological features, it can be divided into four main types: Papillary Thyroid Carcinoma (PTC), Follicular Thyroid Carcinoma (FTC), Medullary Thyroid Carcinoma (MTC), and Anaplastic Thyroid Carcinoma (ATC). PTC is the most common and has a better prognosis, while ATC is the most malignant and has the worst prognosis [141,142]. In addition, ATC progresses rapidly, and most patients have already entered the advanced metastatic stage at the time of initial diagnosis, showing obvious resistance to traditional systemic chemotherapy and targeted therapy [143].

A recent report by Prof. Xiang Qian Zheng's team 2025 elucidates the mechanism by which m^5C modification mediates multidrug resistance (MDR) to ATC by driving alternative splicing reprogramming. Firstly, high expression of m^5C writing enzyme NSUN2 was found to be significantly and positively correlated with drug resistance in patients. Mechanistically, NSUN2 catalyzes the m^5C modification of splicing factor SRSF6 mRNA; ALYREF recognizes this m^5C mark and mediates nuclear export of SRSF6 mRNA, resulting in increased SRSF6 protein translation and accumulation. Accumulated SRSF6 protein reprograms

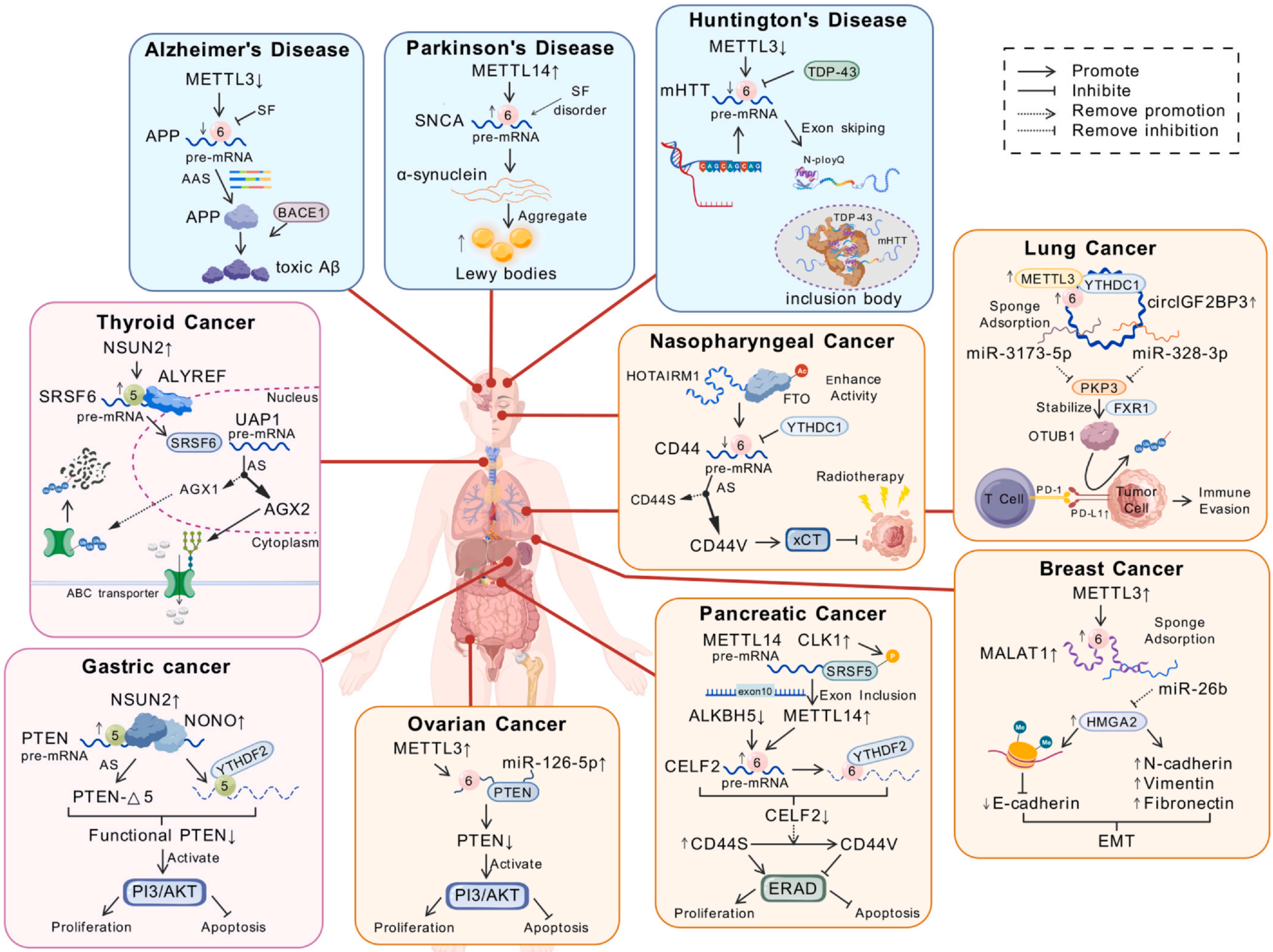

Fig. 3. Aberrant RNA modifications drive disease-specific dysregulation of alternative splicing in cancer and neurological disorders. This schematic summarizes the roles of m^{6}A and m^{5}C RNA methylation in mediating pathogenic alternative splicing events across various cancers and neurological diseases. Each panel illustrates disease- or cancer type–specific mechanisms, highlighting dysregulated RNA-modifying enzymes (writers, erasers, readers), target pre-mRNAs, and splicing factors that contribute to oncogenesis or neurodegeneration. Regulatory interactions are indicated by arrows (activation or promotion, solid arrowhead), blunt lines (inhibition), T-shaped lines (suppression of activation), or dashed lines (indirect regulation), as defined in the accompanying legend. A schematic human body diagram provides anatomical context for the tissue and organ localization of the respective diseases. Figure created using BioGDP.com. [258].

alternative splicing of the UAP1 gene—a key enzyme for UDP-GlcNAc synthesis—prompting a shift from the AGX1 isoform to the AGX2 isoform. Because AGX2 contains an additional 17-amino-acid segment that confers higher substrate-binding activity, it boosts production of the glycosyl donor UDP-GlcNAc and thereby enhances global N-glycosylation. In particular, increased N-glycosylation improves the stability of ABC transporters and accelerates drug efflux, leading to MDR. Application of NSUN2 inhibitors enhances ubiquitinated degradation of ABC transporter proteins and increases intracellular drug concentration (Fig. 3) [144].

RNA modification and alternative splicing play a critical role in the treatment of thyroid cancer, however, the current research on RNA modification and alternative splicing in thyroid cancer still needs to be deepened.

4.1.5. Ovarian cancer

Ovarian cancer (OC) is a highly malignant tumor arising from ovarian tissues and can occur at any age, with the highest mortality rate among all gynecological malignancies. Due to its asymptomatic nature in the early stages, most patients present with advanced disease, characterized by symptoms such as abdominal distension, palpable masses, and ascites. The exact etiology of ovarian cancer remains unclear, although risk factors such as genetic mutations (e.g., BRCA1/BRCA2), hormonal imbalances, and lifestyle factors are believed to contribute to its pathogenesis.

Emerging studies have implicated m^{6}A RNA modification in the development and progression of ovarian cancer. A research group has reported that METTL3, a key m^{6}A methyltransferase, and its downstream target microRNA-126–5p (miR-126–5p) are significantly upregulated in ovarian cancer tissues and are positively correlated with tumor progression. Mechanistically, METTL3 promotes the maturation of miR-126–5p through m^{6}A-dependent regulation. The miR-126–5p subsequently targets and suppresses PTEN, a well-known tumor suppressor gene, thereby activating the PI3K/Akt/mTOR signaling pathway. This activation enhances tumor cell proliferation and inhibits apoptosis. Notably, silencing METTL3 impairs miR-126–5p maturation, restores PTEN expression, and suppresses tumor growth by blocking the PI3K/Akt/mTOR pathway, suggesting a potential therapeutic strategy for ovarian cancer (Fig. 3) [145]. In addition to METTL3, the m^{6}A eraser ALKBH5 has also been found to be overexpressed in ovarian cancer patients and is significantly associated with tumor proliferation and poor clinical outcomes. Elevated ALKBH5 levels lead to reduced m^{6}A methylation near exon 5 of SRSF10, a key splicing factor, resulting in the skipping of this exon (Fig. 3) [146]. The resulting aberrant isoform of SRSF10 may disrupt the normal splicing of its downstream target genes, such as MYB and RFC5, thereby promoting oncogenic processes. These findings underscore the complex regulatory role of m^6A modification in ovarian cancer and highlight novel molecular targets for future therapeutic interventions [147,148].

4.1.6. Nasopharyngeal cancer

Nasopharyngeal cancer (NPC) is a malignant tumor arising from the epithelial lining of the nasopharynx, most commonly located on the lateral walls and roof of the nasopharyngeal cavity. It is a highly prevalent cancer in China, particularly in the southern coastal regions, and represents the most frequently diagnosed malignancy among otorhinolaryngological cancers. NPC exhibits a distinct epidemiological profile, with a male-to-female incidence ratio of approximately 2:1–3:1, and a peak incidence among individuals aged 40–50 years [149]. Common clinical manifestations include nasal obstruction, blood-tinged nasal discharge, aural fullness, and hearing loss. The pathogenesis of NPC is multifactorial, involving Epstein-Barr virus (EBV) infection, genetic predisposition, and environmental risk factors. Recent studies have highlighted the involvement of RNA modifications and alternative splicing in NPC development and treatment resistance. Notably, the long non-coding RNA HOTAIRM1 has been found to be upregulated in radiotherapy-resistant NPC cell lines and clinical tissues. Mechanistically, HOTAIRM1 interacts with the m^6A demethylase FTO, promoting its acetylation modification. This post-translational modification enhances FTO stability by inhibiting its ubiquitin-mediated degradation, thereby increasing its demethylase activity on m^6A-modified CD44 transcripts. This enhanced demethylation reduces the recognition and binding of CD44 mRNA by the m^6A reader protein YTHDC1, leading to altered splicing patterns—specifically, a decrease in the canonical isoform CD44S and an accumulation of the variant isoform CD44V. CD44V has been shown to confer resistance to radiation-induced oxidative stress through the activation of antioxidant pathways such as the xCT system, thereby promoting survival and radio resistance of nasopharyngeal carcinoma cells (Fig. 3) [150]. These findings reveal a novel regulatory axis involving HOTAIRM1–FTO–CD44 signaling in NPC, shedding light on the molecular mechanisms underlying radio resistance and offering potential therapeutic targets for improving clinical outcomes in NPC patients.

4.1.7. Gastric cancer

Gastric cancer (GC) is a malignant tumor arising from the epithelial cells of the gastric mucosa and remains the most prevalent cancer type in China. It exhibits significant regional variation in incidence, with higher rates observed in the northwest and eastern coastal regions compared to southern areas. GC predominantly affects individuals over the age of 50, with a male-to-female incidence ratio of approximately 2:1. Early-stage disease often presents with nonspecific or absent symptoms, while progressive disease is typically associated with abdominal pain, weight loss, and other gastrointestinal disturbances [149]. Risk factors for gastric carcinogenesis include Helicobacter pylori infection, dietary habits, and genetic predisposition. Emerging evidence highlights the involvement of RNA modifications and alternative splicing dysregulation in gastric cancer development.

Recent studies have identified the RNA-binding protein NONO as significantly upregulated in gastric cancer tissues. Mechanistically, NONO recruits the m^5C methyltransferase NSUN2 to PTEN mRNA, where they act synergistically to modulate its post-transcriptional fate [151]. On one hand, m^5C methylation alters the secondary structure of PTEN mRNA, exposing cryptic splice sites and promoting the generation of truncated, nonfunctional PTEN isoforms, such as PTEN-Δ5. On the other hand, m^5C modification facilitates the recruitment of m^5C reader proteins, including YTHDF2, which enhances the degradation of PTEN mRNA. Collectively, these processes lead to a marked reduction in functional PTEN protein levels. The loss of PTEN results in constitutive activation of the PI3K/AKT signaling pathway, which drives tumor cell proliferation, inhibits apoptosis, and contributes to the malignant progression of gastric cancer (Fig. 3) [152,153]. These findings reveal a critical role for RNA modification-mediated dysregulation in gastric tumorigenesis and suggest potential molecular targets for therapeutic intervention.

4.2. Convergent oncogenic axis: aberrant RNA modification–splicing reprogramming across cancers

Across diverse cancer types, aberrant RNA-modification patterns have converged on a unified pathogenic principle: the epitranscriptomic rewiring of alternative splicing. Writers, erasers and readers of the m^6A, m^5C and related modifications—most frequently METTL3, ALKBH5, NSUN2 and YTHDC1—are recurrently dysregulated in virtually every tumor lineage. By altering the abundance of these modifications, the enzymes reprogram the expression, subcellular localization or splice-site selection of core SR and SR-like factors (e.g., SRSF5, SRSF6, SRSF10), thereby generating gain-of-function isoforms of driver oncogenes or eliminating tumor-suppressor transcripts.

In nasopharyngeal and pancreatic cancers, elevated m^6A levels enhance YTHDC1 binding to pre-mRNAs, converging on a switch in CD44 splicing that ultimately underlies radio-resistance and oncogenic-pathway activation through lineage-specific downstream mechanisms. Similarly, in gastric and thyroid cancers, an NSUN2/m^5C axis drives the production of pro-oncogenic splice variants that increase drug efflux and activate PI3K/AKT signaling; in ovarian cancer, a METTL3/miR-26–5p/PTEN circuit likewise activates PI3K/AKT to propel tumor progression. In non-small-cell lung cancer, an m^6A-driven METTL3/circIGF2BP3/PKP3/OTUB1 axis licenses immune evasion, whereas in breast cancer a METTL3/MALAT1/miR-26b route promotes EMT. Irrespective of tumor origin, the same "modification-splicing-function axis" repeatedly hijacks canonical signaling cascades—PI3K/AKT, PD-L1/PD-1, EMT and ERAD among others–by manufacturing isoforms with altered scaffolding, subcellular localization or degradation properties.

This recurrent epitranscriptomic logic underscores the universal role of RNA modifications in malignant transformation and provides a compelling rationale for developing pan-cancer therapeutics that target the enzymatic apparatus of RNA marking or the splice variants it generates.

4.3. RNA modification and alternative splicing abnormalities in neurodegenerative diseases

Neurodegenerative diseases constitute a heterogeneous group of disorders characterized by the progressive loss of neuronal structure and function. This category includes well-known conditions such as Alzheimer's disease, Parkinson's disease, and Huntington's disease, among others. These disorders profoundly impair cognitive abilities, motor function, and overall quality of life, placing a substantial burden on patients, families, and healthcare systems. Emerging evidence suggests that aberrant RNA modifications and dysregulated alternative splicing play pivotal roles in the pathogenesis of neurodegenerative diseases. Accumulating studies have highlighted the involvement of these post-transcriptional regulatory mechanisms in disease onset and progression. A deeper understanding of their functional interplay may provide critical insights into the underlying molecular mechanisms of neurodegeneration and offer novel therapeutic targets for the development of effective interventions.

4.3.1. Alzheimer's disease

Alzheimer's disease (AD) is the most prevalent neurodegenerative disorder and is characterized by progressive cognitive decline and memory impairment. Its major pathological hallmarks include the extracellular deposition of amyloid-β (Aβ) peptides into senile plaques, intracellular accumulation of hyperphosphorylated Tau protein into neurofibrillary tangles (NFTs), and widespread neuronal loss.

Emerging evidence highlights a critical role for m^6A RNA modification in the progression of AD. The amyloid precursor protein (APP) gene, which encodes the precursor for Aβ production, undergoes aberrant alternative splicing in AD. Studies have shown that the m^6A methyltransferase METTL3 is abnormally accumulated in the brains of AD patients, leading to a reduction in m^6A modification of APP mRNA [154]. This altered m^6A landscape disrupts the binding of splicing factors to APP pre-mRNA, resulting in aberrant splicing and increased production of neurotoxic Aβ isoforms [155]. These Aβ fragments progressively accumulate in the brain, forming senile plaques, which trigger neuroinflammatory responses and neuronal damage, ultimately contributing to cognitive decline.

Another key gene implicated in AD pathogenesis is Beta-site Amyloid Precursor Protein Cleaving Enzyme 1 (BACE1), which encodes β-secretase, an enzyme critical for the cleavage of APP to generate Aβ. Notably, studies have demonstrated that METTL3 deletion alleviates Aβ-induced pathology in mouse models of AD. This suggests that m^6A methylation may influence AD progression by modulating BACE1 mRNA stability or translation, thereby affecting Aβ production (Fig. 3) [156].

Beyond the APP and BACE1 pathways, m^6A modification has also been implicated in the dysregulation of alternative splicing of other genes associated with synaptic function and neuronal signaling. For instance, Neurexin 1 (NRXN1), a gene essential for synapse formation and function, exhibits m^6A-dependent splicing abnormalities in AD [157].

These splicing alterations may compromise synaptic structure and function, impair neuronal communication and contribute to cognitive deficits. Collectively, these findings underscore the significance of m^6A RNA modification and its impact on alternative splicing in the molecular pathogenesis of Alzheimer's disease, highlighting novel regulatory mechanisms that may serve as potential therapeutic targets.

4.3.2. Parkinson's disease

Parkinson's disease (PD) is the second most prevalent neurodegenerative disorder, characterized by the selective degeneration of dopaminergic neurons in the substantia nigra pars compacta of the midbrain, along with the intracellular accumulation of aggregated alpha-synuclein into Lewy bodies. These pathological changes give rise to the classical motor symptoms of PD, including bradykinesia, resting tremor, and rigidity.

Emerging evidence highlights the involvement of RNA modifications in the dysregulation of alternative splicing of genes associated with PD. In postmortem brain tissues from PD patients, altered binding of splicing factors near m^6A modification sites has been observed on SNCA mRNA, which encodes alpha-synuclein, when compared to healthy controls [158]. This aberrant splicing contributes to the generation of alpha-synuclein isoforms with altered structural and functional properties. Furthermore, studies have demonstrated that overexpression of METTL14, a core component of the m^6A methyltransferase complex, significantly increases m^6A modification of alpha-synuclein mRNA, leading to reduced transcript stability [159]. Notably, alternatively spliced isoforms of alpha-synuclein, such as those forming heterodimers, exhibit increased propensity for aggregation and are more likely to contribute to the formation of Lewy bodies [160]. These pathological aggregates are believed to play a central role in the progressive loss of dopaminergic neurons, ultimately driving disease progression (Fig. 3).

4.3.3. Huntington's disease

Huntington's disease (HD) is an autosomal dominant neurodegenerative disorder caused by an abnormal expansion of CAG trinucleotide repeats ($>$ 35 repeats) in the HTT gene, which encodes the huntingtin protein. The mutant huntingtin protein (mHTT) contains an elongated polyglutamine (polyQ) tract that leads to protein misfolding and the formation of insoluble aggregates in the nucleus or cytoplasm of neurons. These aggregates are a hallmark of HD pathology and are closely associated with progressive neuronal degeneration [161].

TAR DNA-binding protein 43 (TDP-43) is a ubiquitously expressed RNA/DNA-binding protein that is highly enriched in neurons, where it plays essential roles in RNA splicing, transcription, and mRNA stability. TDP-43 regulates the splicing of pre-mRNA transcripts involved in neuronal development by binding to UG-rich sequences, thereby ensuring accurate exon inclusion or exclusion. It has also been identified as a pathological substrate in multiple neurodegenerative diseases, including amyotrophic lateral sclerosis (ALS) [162−164]. Recent studies have shown that phosphorylated TDP-43 accumulates in the cytoplasm of both HD mouse models and postmortem human HD brains [165]. Notably, mHTT aggregates can sequester RNA-binding proteins, including TDP-43, into inclusion bodies, leading to their nuclear depletion. This mislocalization reduces TDP-43 binding to its target RNAs, particularly those associated with HD-related differentially expressed genes, thereby disrupting normal RNA splicing. In parallel, dysfunction of METTL3, a key component of the m^6A methyltransferase complex, has been observed in HD models. This results in a global reduction in m^6A modification levels near mutant HTT transcripts. Given that TDP-43 preferentially binds to m^6A-modified RNAs, the loss of m^6A marks further impairs TDP-43–RNA interactions. This dysregulation leads to aberrant exon skipping and the generation of dysfunctional protein isoforms in several HD-related genes, including mutant HTT itself, thereby exacerbating neurodegeneration (Fig. 3) [166].

Collectively, these findings suggest that the cytoplasmic mislocalization of TDP-43, its co-localization with mHTT aggregates, and alterations in m^6A RNA modification levels synergistically contribute to widespread RNA splicing abnormalities in HD. These insights provide a novel mechanistic understanding of HD pathogenesis and highlight potential therapeutic targets for future intervention strategies aimed at restoring RNA homeostasis and improving patient outcomes.

4.4. Convergent pathogenic axis: aberrant RNA modification–splicing reprogramming across neurodegenerative diseases

Across major neurodegenerative disorders, aberrant RNA modifications and the consequent disruption of alternative splicing constitute a deeply conserved pathogenic framework. Although Alzheimer's disease, Parkinson's disease and Huntington's disease differ in clinical phenotype and neuropathology, they converge on a unifying molecular principle: disruption of m^6A and related epitranscriptomic marks skews splice-factor activity or directly remodels disease-critical pre-mRNAs, precipitating global splicing imbalance and driving neurodegeneration.

In AD, METTL3-mediated hyper-methylation biases APP and BACE1 splicing toward amyloidogenic isoforms, amplifying Aβ generation; in PD, altered m^6A stoichiometry destabilizes SNCA (α-synuclein) transcripts and favors aggregation-prone splice variants, which aggregate into Lewy bodies; in HD, a global decline in m^6A abundance impairs TDP-43-RNA interactions to corrupt HTT and downstream transcript splicing, accelerating neuronal demise. Regardless of disease context, the recurrent "RNA-modification/splice reprogramming/toxic protein production or aggregation" axis positions epitranscriptomic dysregulation as a central, cross-cutting driver of neurodegeneration.

These insights not only unify the molecular etiology of seemingly disparate diseases but also provide a compelling rationale for broad-spectrum therapeutic strategies that target RNA-modifying enzymes or their pathogenic splice outputs.

4.5. RNA modification and alternative splicing abnormalities in other diseases

Autoimmune diseases (AIDs) are a group of disorders characterized by the immune system's aberrant response to self-antigens, leading to the production of autoantibodies and the activation of sensitized lymphocytes that attack the body's own tissues, resulting in chronic inflammation, tissue damage, and organ dysfunction. In recent years, dysregulation of RNA modifications and their impact on alternative splicing have emerged as a key mechanism in the pathogenesis of autoimmune diseases. Studies have revealed that METTL3/ALKBH5-controlled m^6A modification directly targets the 3′UTR of SLC15A3 mRNA, enhancing its stability and translational efficiency and thereby elevating SLC15A3 protein levels. Increased SLC15A3 facilitates recruitment of TASL (Toll-like receptor adaptor interacting with SLC15A3) to lysosomes, where TASL subsequently activates IRF5 (interferon regulatory factor 5). This cascade drives transcription of downstream inflammatory genes, promotes M1 macrophage polarization, and exacerbates psoriatic inflammation [167,168]. Additionally, the expression of a truncated splice variant of CTLA-4 (1/4 CTLA-4), which lacks the ligand-binding domain necessary for inhibiting T cell activation, is elevated in SLE patients. This isoform fails to suppress T cell responses, thereby contributing to the breakdown of immune tolerance [169].

In infectious disease research, a seminal 2019 study established that the host methyltransferase NSUN2 installs internal m^5C modifications on HIV-1 pre-mRNA within the nucleus. Mechanistically, the modification operates as a molecular platform: by engaging the m^5C reader MBD2, it recruits and modulates SR and hnRNPs splicing factors, thereby refining splice-site selection and markedly increasing the abundance of regulatory viral transcripts such as tat and rev. Concomitantly, the m^5C mark enhances ribosome loading on HIV-1 mRNAs, accelerating translation of the structural proteins Gag and Pol by two- to three-fold [170]. Collectively, these findings position NSUN2-mediated m^5C modification as a critical regulatory layer of HIV-1 gene expression, advance our understanding of the viral epitranscriptome, and provide a theoretical framework for therapeutic strategies targeting RNA-modification pathways.

In the reproductive system, ALKBH5-dependent m^6A demethylation has been shown to play a critical role during spermatogenesis, particularly in the meiotic and haploid stages. ALKBH5 binds to the long 3′-UTR regions of key transcripts such as Pou5f1 and Id4, regulating their mRNA stability and expression through m^6A demethylation. This regulatory mechanism is essential for the proper development of male germ cells [171].

These examples illustrate the broad and diverse impact of RNA modifications and alternative splicing dysregulation across a range of diseases beyond cancer and neurodegeneration, highlighting their relevance as potential therapeutic targets and biomarkers.

5. Disease diagnosis and treatment strategies based on RNA modifications and alternative splicing

5.1. Disease diagnosis

5.1.1. Biomarker discovery

RNA modifications and alterations in alternative splicing have emerged as promising sources of biomarkers for the early detection, diagnosis, and prognosis of various diseases. Among RNA modifications, m^6A has attracted particular attention due to its critical regulatory roles in gene expression and its dysregulation in pathological conditions, especially cancer. Aberrant expression of m^6A regulatory enzymes, such as the methyltransferase METTL3, has been frequently observed in multiple tumor types and is often associated with tumor initiation, progression, and poor clinical outcomes [172]. The detection of the expression levels of these m^6A regulatory enzymes in serum or tumor tissues, as well as the m^6A modification status of its downstream oncogenic transcripts, holds significant potential as a biomarker for early cancer diagnosis and prognostic evaluation [173−176]. Furthermore, distinct m^6A modification patterns have been identified in tumor tissues compared to normal tissues, with certain differentially modified sites showing promise as highly specific and sensitive biomarkers [177]. In neurodegenerative diseases such as AD, dysregulation of RNA modifications and alternative splicing has also been widely reported. Studies have demonstrated that m^6A modification levels at key disease-associated genes are altered in the brains of AD patients, which in turn affects the alternative splicing of these transcripts and leads to the production of pathogenic protein isoforms [178−180]. The measurement of m^6A modification levels and alternative splicing variants of AD-related genes—such as the APP gene—in brain tissue or cerebrospinal fluid (CSF) may offer valuable insights for early diagnosis and disease monitoring. Moreover, specific splicing events have been proposed as potential biomarkers for AD diagnosis.

From a clinical perspective, biomarkers based on RNA modifications and alternative splicing offer several advantages, including high specificity and sensitivity. These biomarkers enable the detection of molecular-level changes at early stages of disease, thereby providing opportunities for timely intervention. Compared to traditional diagnostic markers, RNA-based biomarkers more accurately reflect the underlying pathogenic mechanisms, potentially improving diagnostic accuracy and reliability [181].

However, despite their promise, these biomarkers require further validation through large-scale clinical studies to assess their reproducibility, stability, and applicability across diverse populations and disease subtypes. Such efforts will be critical for translating these findings into routine clinical practice.

5.1.2. Development of diagnostic technologies

With the rapid advancement of biotechnological tools, methods for detecting RNA modifications and alternative splicing events—such as high-throughput sequencing, PCR-based assays, and other emerging technologies—have been increasingly applied in the field of disease diagnosis.

High-throughput RNA sequencing (RNA-seq) has become a cornerstone technology for the comprehensive and precise identification of RNA modifications and alternative splicing patterns [182]. By analyzing RNA-seq data, researchers can detect disease-associated differential modification sites and aberrant splicing isoforms. For instance, emerging evidence reveals that small RNAs such as miRNAs, PIWI-interacting RNAs (piRNAs) and tRNA-derived small RNAs (tsRNAs) exhibit diverse RNA modifications, and high-resolution, high-throughput detection methods are now uncovering their potential as sensitive disease markers [183]. Reverse transcription quantitative PCR (RT-qPCR) is widely used to measure the expression levels of specific RNA isoforms or modified transcripts [184]. By designing isoform-specific primers, RT-PCR can accurately reflect changes in alternative splicing with high sensitivity and reproducibility, making aberrant splicing isoforms suitable as markers for cancer and potential targets for cancer therapy [185]. Beyond conventional methods, emerging technologies are expanding the toolkit for RNA modification and splicing analysis [186,187]. Mass spectrometry-based approaches allow direct detection of RNA chemical structures and modification sites, providing high-resolution insights into RNA modification patterns [188,189]. Nanopore sequencing, with its long-read capability and real-time detection, offers a powerful platform for identifying RNA modifications at the single-molecule level. Here, Nano-tRNAseq, which utilizes nanopore technology, sequences native tRNA populations in a single experiment, offering both quantitative measurements of tRNA levels and insights into the dynamics of tRNA modifications [190,191].

The continuous development and refinement of these advanced technologies are expected to significantly enhance the clinical application of RNA modification and alternative splicing profiling in disease diagnosis. These innovations hold great promise for enabling earlier detection, more accurate diagnosis, and the realization of personalized medicine tailored to individual RNA-based molecular signatures.

5.2. Treatment strategies

5.2.1. Targeting RNA-modifying enzymes as a therapeutic strategy

Targeting RNA-modifying enzymes represents a promising therapeutic approach for correcting aberrant RNA modifications and restoring normal alternative splicing patterns in the context of disease treatment. These enzymes, which include writers, erasers, and readers of RNA modifications, play essential roles in regulating RNA stability, localization, translation, and splicing. Dysregulation of their expression or activity has been implicated in the disruption of RNA modification landscapes, leading to aberrant splicing events and the progression of various diseases. Therefore, modulating the function of RNA-modifying enzymes offers a novel and potentially effective strategy for restoring RNA homeostasis and ameliorating disease phenotypes.

Among RNA modifications, m^6A has been extensively studied in cancer due to its critical regulatory roles in oncogenic processes. METTL3, a core component of the m^6A methyltransferase complex, is frequently overexpressed in multiple tumor types. This aberrant upregulation enhances m^6A modification levels on oncogenic transcripts, increasing their stability and translational efficiency, thereby promoting tumor cell proliferation and metastasis. In particular, METTL3 also promotes tumor progression in acute myeloid leukemia (AML) by regulating the translation of genes such as c-MYC and BCL-2 [192]. To counteract this mechanism, efforts have been directed toward the development of METTL3 inhibitors that can suppress the pathological elevation of m^6A levels and inhibit tumor progression [193,194]. Recent progress in developing small molecule inhibitors aimed at specific writers or erasers to reverse the epitranscriptome changes in cancer cells and their potential therapeutic applications [195,196]. Several small-molecule inhibitors, such as STM2457, which specifically targets and inhibits METTL3 activity. In mouse in vivo experiments, it significantly suppresses the proliferation of AML cells and tumor growth without apparent toxic side effects [197]. These compounds specifically target the catalytic site of METTL3, blocking its methyltransferase activity and reducing m^6A levels in tumor cells. Consequently, this leads to decreased expression of oncogenic transcripts, ultimately suppressing tumor progression. Moreover, these inhibitors are being evaluated for their pharmacokinetic properties and potential synergistic effects when combined with conventional chemotherapeutic agents or targeted therapies, which may enhance treatment efficacy and overcome drug resistance.

Despite the therapeutic promise, targeting RNA-modifying enzymes presents several challenges. First, these enzymes are involved in a wide range of physiological processes, and their inhibition may lead to off-target effects that disrupt normal cellular functions. For example, while METTL3 inhibitors effectively suppress tumor growth, they may also interfere with the differentiation and development of normal tissues. Second, RNA-modifying enzymes often function within complex regulatory networks, and perturbation of a single enzyme may not be sufficient to fully restore RNA homeostasis.

Therefore, a deeper understanding of the in vivo mechanisms and interactions of RNA-modifying enzymes is essential to enhance the specificity and efficacy of therapeutic interventions while minimizing adverse effects. Future research should focus on the development of tissue-specific delivery systems and selective inhibitors with improved targeting precision.

Recent advances in RNA engineering have introduced novel tools for precise manipulation of RNA modifications. One such innovation is the Targeted RNA Methylation (TRM) system, an engineered platform that enables site-specific installation of m^6A modifications on endogenous transcripts. By fusing dCas13 with methyltransferases (such as METTL3/METTL14), this system allows for the controlled induction of m^6A-mediated changes in RNA abundance and alternative splicing, providing a powerful tool for dissecting the functional consequences of individual RNA modifications [198]. Conversely, a study has developed a targeted m^6A demethylation system (TRME) by fusing the catalytic domain of dCas13a with that of ALKBH5, achieving precise and reversible demethylation of m^6A sites, and using it to regulate the differentiation of human pluripotent stem cells (hESCs) [199,200]. The TRM and TRME system hold great potential for both basic research and therapeutic applications, offering a new avenue for targeted epitranscriptome editing in the future. Based on this, significant progress has been made in clinical research for certain diseases, such as bladder cancer [201]. More significantly, systems targeting m^5C modification have also been developed [202−204].

5.2.2. Modulating alternative splicing as a therapeutic strategy

Targeting alternative splicing through antisense oligonucleotides (ASOs) and small molecule compounds has emerged as a promising therapeutic strategy, grounded in the mechanistic understanding of splicing regulation [205−208]. This approach enables precise post-transcriptional control of gene expression and holds significant potential for the treatment of various genetic and acquired diseases [209].

Antisense oligonucleotides are short, synthetic, single-stranded nucleic acid sequences, typically 15–22 nucleotides in length, designed to specifically bind to target pre-mRNA through complementary base pairing. In the context of alternative splicing, ASOs are engineered to hybridize to splice sites or regulatory elements at exon–intron boundaries or within intronic splicing enhancers/silencers. Upon binding, ASOs interfere with the recognition of these sites by the spliceosome, thereby modulating splice site selection and promoting the inclusion or exclusion of specific exons during mRNA maturation. This enables the functional study and therapeutic manipulation of alternative splicing events [206,210,211].

For instance, an ASO that switch PKM splicing from the cancer-associated PKM2 isoform to the PKM1 isoform, in genetic hepatocellular carcinoma (HCC) mouse models, induces Pkm splicing switching and inhibit tumorigenesis successfully [212]. Furthermore, it has been proven that ASO-mediated skipping of cancer-driving exons is a precision-therapy strategy for diseases such as Duchenne muscular dystrophy and Spinal muscular atrophy (SMA) [213−215]. These studies establish ASOs as powerful tools for modulating mRNA splicing.

In addition to ASOs, small molecule compounds represent an alternative and pharmacologically attractive approach for modulating alternative splicing. These molecules can interact with splicing factors or spliceosomal components, altering their conformation, activity, or RNA-binding properties, thereby influencing spliceosome assembly and splice site selection [205,208].

One such compound, E7107, targeting the U2 snRNP subunit SF3b, inhibiting its interaction with target mRNAs and thereby altering the splicing patterns of oncogenic transcripts [216,217]. This compound has demonstrated remarkable efficacy in both solid-tumor and leukemia models [218−220]. Remarkably, H3B-8800—an orally available small-molecule splicing modulator—selectively eliminates spliceosome-mutant cancer cells by targeting the SF3b complex [221]. H3B-8800 has already shown high therapeutic potential in early-phase clinical trials [222]. These findings underscore the therapeutic potential of small molecule splicing modulators in oncology.

Despite their therapeutic promise, the clinical application of alternative splicing-targeted therapies faces several challenges. For ASOs, in vivo stability and efficient delivery remain major limitations [223]. Unmodified ASOs are highly susceptible to nuclease degradation, which hampers their accumulation at target sites and limits therapeutic efficacy. To overcome this, chemical modifications such as 2′-O-Me are commonly introduced to enhance nuclease resistance, improve binding affinity, and prolong in vivo half-life [224]. In terms of delivery, effective strategies are required to ensure targeted accumulation and cellular uptake of ASOs. Commonly used delivery systems include liposome-based carriers and nanoparticles [225]. However, these approaches often suffer from limited tissue specificity and suboptimal bioavailability, necessitating further optimization.

Similarly, the development of small molecule splicing modulators presents challenges in compound screening, optimization, and selectivity. High-throughput screening and computational modeling are essential tools in the discovery and refinement of lead compounds [226,227]. However, off-target effects on normal splicing events remain a concern, emphasizing the need for improved specificity to minimize unintended disruptions of physiological RNA processing.

5.2.3. Gene therapy approaches

Gene therapy has emerged as a promising therapeutic approach for treating diseases caused by genetic mutations over the past decade. Initially, gene therapy focused on knocking out or repairing pathogenic genes at the DNA level, with β-thalassemia as a paradigmatic example: a typical therapeutic strategy involves the delivery of the CRISPR/Cas9 system into the patient's hematopoietic stem cells, followed by ex vivo gene editing and subsequent reinfusion of the corrected cells back into the patient [228,229]. Subsequent innovations further refine CRISPR/Cas9 to precisely target the erythroid-specific BCL11A enhancer, restoring the production of fetal hemoglobin [230,231]. This approach holds the potential for long-term or even curative treatment of β-thalassemia.

With the rapid advancement of gene editing technologies, gene therapy offers new opportunities for addressing a wide range of disorders associated with RNA dysregulation, which caused by aberrant RNA modification and dysfunctional alternative splicing.

In contrast, CRISPR/Cas9 technology offers a more transformative approach to modulating alternative splicing, with current efforts falling into three main strategies. First, pathogenic exons are directly excised to prevent aberrant protein production and achieve therapeutic benefit [232 − 235]. Second, CRISPR-Cas9 cytidine and adenosine base editors are used to mutate specific bases within splice-acceptor sites at the genomic level, thereby programming exon skipping and restoring normal gene expression [236 − 238]. Third, when a disease-linked mutation disrupts splicing, the offending nucleotide is corrected in situ, re-establishing physiological isoform patterns and curing the disorder [239]. Recently, a powerful tool—the dCasRx-RBM25—has been developed. Leveraging gRNA arrays for combinatorial targeting, it can activate or repress selected exons at the RNA level, laying the groundwork for splicing-directed therapeutics [240]. Coupled with high-throughput sequencing and other bioinformatic pipelines to identify critical exons, this ensures precise and efficient splice-pattern modulation [234,241 − 243].

Beyond splicing, TRM and TRME systems represent CRISPR-based applications in RNA modification. By fusing dCas13 to m^{6}A writers or erasers, these platforms enable site-specific methylation editing on RNA, thereby tuning gene expression with nucleotide precision [198 − 200,204].

Despite its therapeutic potential, gene therapy faces several challenges that must be addressed to ensure its safety and efficacy. Off-target effects represent a major safety concern in gene editing, as unintended genomic modifications induced by CRISPR/Cas9 may lead to deleterious consequences, including potential oncogenic risks. Therefore, improving the specificity of gene editing tools and minimizing off-target activity remain urgent priorities in the field [232].

In addition, the delivery systems used in gene therapy require further optimization. Commonly used viral vectors, such as adeno-associated virus (AAV), are limited by their small cargo capacity and potential immunogenicity, which can restrict their applicability and effectiveness [244]. The development of novel delivery platforms—-such as engineered viral vectors or advanced non-viral delivery systems—is a key direction for future research aimed at enhancing both the safety and efficiency of gene therapy [245].

Moreover, the ethical and societal implications of gene editing warrant careful consideration. The application of gene editing technologies, particularly in the context of germline editing, raises complex ethical questions. Robust ethical frameworks and regulatory policies must be established and enforced to ensure responsible and equitable use of gene therapy in clinical settings.

6. Research summary and future perspectives

6.1. Mechanistic summary

In this study, we have systematically elucidated the molecular mechanisms by which RNA modifications contribute to aberrant alternative splicing and subsequently lead to disease pathogenesis. RNA modifications represent a crucial layer of post-transcriptional regulation and encompass a diverse array of chemical alterations, including but not limited to m^{6}A, m^{5}C, m^{1}A, and m^{7}G. These modifications can influence alternative splicing through multiple regulatory pathways, thereby playing pivotal roles in the control of gene expression.

In the context of oncology, dysregulation of m^{6}A modification—mediated by its writer enzymes such as METTL3 and METTL14—has been implicated in the disruption of normal splicing patterns. At the molecular level, m^{6}A modifications can modulate the interaction between mRNA and splicing factors by recruiting specific reader proteins, such as members of the YTH domain-containing family. This influences spliceosome assembly and splice site selection. For instance, YTHDC1 has been shown to recognize m^{6}A-modified transcripts and facilitate the recruitment of splicing factor SRSF3, thereby promoting exon inclusion. In contrast, the loss of m^{6}A at specific sites favors the binding of SRSF10, which promotes exon skipping. Such alterations in splicing regulation can result in the aberrant expression of oncogenes and contribute to tumorigenesis.

RNA modifications also play a significant role in neuropsychiatric disorders. In Alzheimer's disease, for example, abnormal m^{6}A modification has been linked to the dysregulation of alternative splicing in key disease-associated genes such as APP. Altered m^{6}A levels have been shown to give rise to aberrant splicing isoforms, which in turn affect protein function and metabolism, ultimately leading to neuronal dysfunction and cell death. Furthermore, m^{6}A modifications may contribute to neurodegenerative and psychiatric conditions by modulating the binding of RNA-binding proteins to mRNA and disrupting the normal regulation of alternative splicing.

Taken together, our findings highlight the critical role of RNA modifications in disease development through their regulatory effects on alternative splicing. RNA modifications function as fine-tuned "molecular switches" that precisely modulate alternative splicing at multiple levels—epigenetically, through interactions with RNA-binding proteins, and via cellular stress response pathways–by altering RNA structure and functionality. Disruptions in this finely balanced regulatory system can lead to widespread splicing abnormalities, ultimately contributing to a range of pathological conditions.

These insights not only offer novel perspectives on disease pathogenesis but also provide a robust theoretical foundation for the development of early diagnostic strategies, precision therapeutic interventions, and improved prognosis evaluation in related diseases.

6.2. Future perspectives: clinical translation and therapeutic potential

Targeting the RNA modification–alternative splicing axis represents a promising frontier in precision medicine, offering the potential for personalized therapeutic strategies tailored to the molecular pathology of individual diseases. By elucidating the specific mechanisms through which RNA modifications and aberrant splicing contribute to disease pathogenesis, key regulatory enzymes, splicing factors, and disease-associated splicing events can be identified. This knowledge lays a critical foundation for the development of highly specific therapeutic agents and targeted intervention strategies.

In the context of oncology, the development of small-molecule inhibitors targeting RNA-modifying enzymes—such as METTL3—holds promise for modulating aberrant m^{6}A modifications and restoring normal splicing patterns in a tumor-specific manner. These compounds can selectively regulate gene expression in malignant cells, thereby suppressing tumor growth and metastasis. Such precision-targeting approaches have the potential to minimize off-target toxicity to normal tissues, enhance therapeutic efficacy, and improve both survival rates and quality of life for cancer patients.

Despite these promising developments, the clinical translation of RNA modification- and splicing-targeted therapies faces several key challenges. One of the most critical issues is tissue specificity. Given the marked heterogeneity in RNA modification patterns and splicing profiles across different tissues and cell types, the development of tissue-specific delivery systems is essential for effective and safe therapeutic application. Without precise targeting, therapeutic agents may fail to reach the intended disease site or may induce unintended effects in non-target tissues. Recent studies demonstrate that lipid nanoparticles (LNPs) can deliver antibodies or nucleic-acid drugs to specific organs with unprecedented precision. SORT LNPs, for instance, redirect base editors to liver or lung, enabling on-site mutation correction at therapeutically relevant efficiencies [246,247]. Likewise, biodegradable polymers such as PLGA, PLA, and PCL are fabricated into polymeric nanoparticles (PNPs) that encapsulate therapeutics and deliver them selectively to target organs [248,249].

Another major hurdle is off-target liability. RNA-modifying enzymes and splicing factors belong to highly conserved families with multiple paralogues that share overlapping substrate motifs. Small-molecule inhibitors directed against one splice regulator frequently engage structurally related proteins, thereby distorting the splicing of hundreds to thousands of transcripts. To mitigate this, recent studies have highlighted how artificial intelligence and machine learning enhance RNA and protein structure prediction and ligand-screening efficiency [250,251]. When integrated with multi-omics data, including transcriptome-wide splicing profiling and epitranscriptomic mapping, these multimodal datasets enable a comprehensive assessment of on-target versus off-target effects of candidate compounds and offer new therapeutic strategies for diverse diseases. Additionally, gene editing aimed at specific writers, erasers, or readers carries the latent oncogenic risk of off-target genomic alterations. Random mutagenesis of the REC3 domain has been exploited to isolate high-fidelity Cas9 variants [252]. Genome-wide off-target detection platforms such as GUIDE-seq, Digenome-seq, and CIRCLE-seq are then used to systematically evaluate editing products, provide empirical evidence for sgRNA design, and, in conjunction with machine-learning models, further optimize specificity [253−256].

Looking ahead, the field of RNA modification and alternative splicing is poised for significant breakthroughs driven by technological advancements. The development and application of single-cell sequencing technologies will enable high-resolution profiling of RNA modifications and splicing dynamics at the single-cell level, shedding light on how cellular heterogeneity influences disease progression. Concurrently, the integration of artificial intelligence (AI) and machine learning tools into biomedical research will facilitate the analysis of large-scale omics datasets, uncovering novel associations between RNA regulatory mechanisms and disease phenotypes [257]. These approaches will accelerate the discovery of diagnostic biomarkers and the identification of novel drug targets.

Furthermore, continuous improvements in gene editing technologies, particularly the refinement of CRISPR-Cas systems, are expected to provide powerful tools for correcting disease-associated RNA modifications and splicing defects at the genomic level. These innovations hold the potential to transform both the diagnosis and treatment of a wide range of diseases, significantly improving diagnostic accuracy, therapeutic precision, and patient outcomes.

In summary, as our understanding of RNA biology deepens and technological capabilities expand, the clinical translation of RNA modification- and alternative splicing-targeted therapies will likely usher in a new era of precision medicine, offering novel therapeutic hope for patients with currently untreatable or poorly managed conditions.

CRediT authorship contribution statement

Shu-Juan Xie: Visualization, Project administration. **Zhen-Dong Xiao:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Jin Zeng:** Writing – original draft, Visualization, Conceptualization. **Li-Ting Diao:** Visualization, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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